HANDS-ON WRITING A DMP

Faculteit Geneeskunde en Gezondheidswetenschappen

GHENT UNIVERSITY

Slides: https://osf.io/2u7k3

CC BY

Attribute to: Ghent University Data Stewards

DOZA

UNIVERSITY LIBRARY / UNIVERSITEITSBIBLIOTHEEK

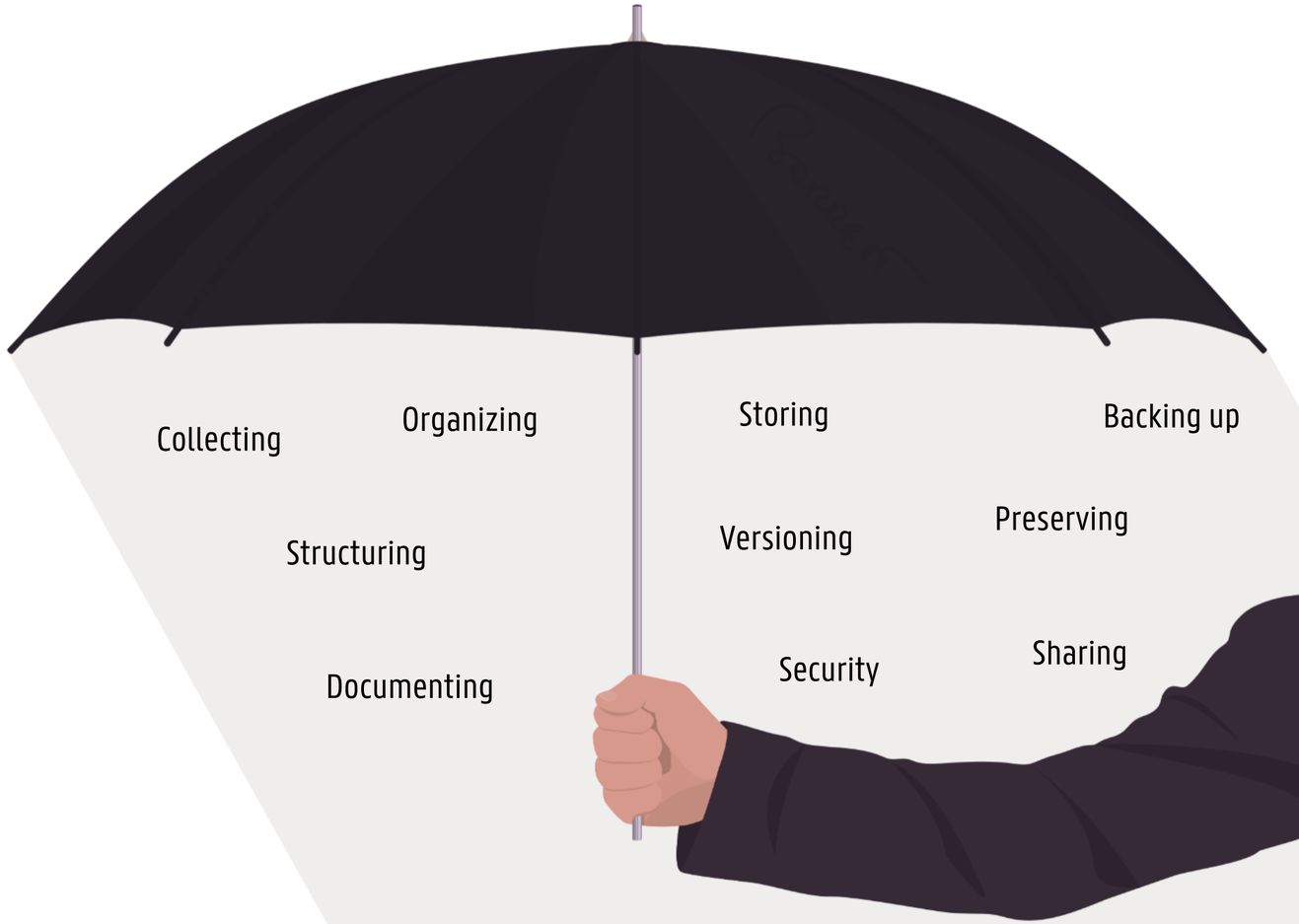
RDM AND DMP POLICIES

GHENT UNIVERSITY

WHAT IS RESEARCH DATA MANAGEMENT (RDM)?

- Actions to **take care of data during the course of research and beyond**.
- An integral part of research.

"the compilation of many small practices that make your data easier to understand, less likely to be lost, and more likely to be usable during a project or ten years later" (Briney 2015)



GHENT UNIVERSITY

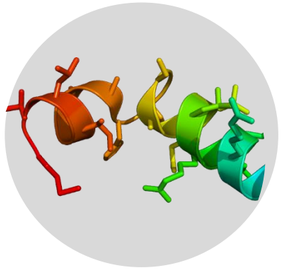
Adapted from 'RDM: An Overview' by Research Support Team, IT Services (University of Oxford), licensed under CC-BY-NC-SA 4.0 and OpenClipart CC BY

RESEARCH DATA CAN MEAN A LOT

Survey



Measurement



Observation



Code



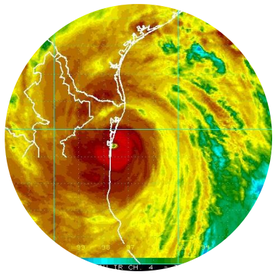
Image



Document



Visualisation

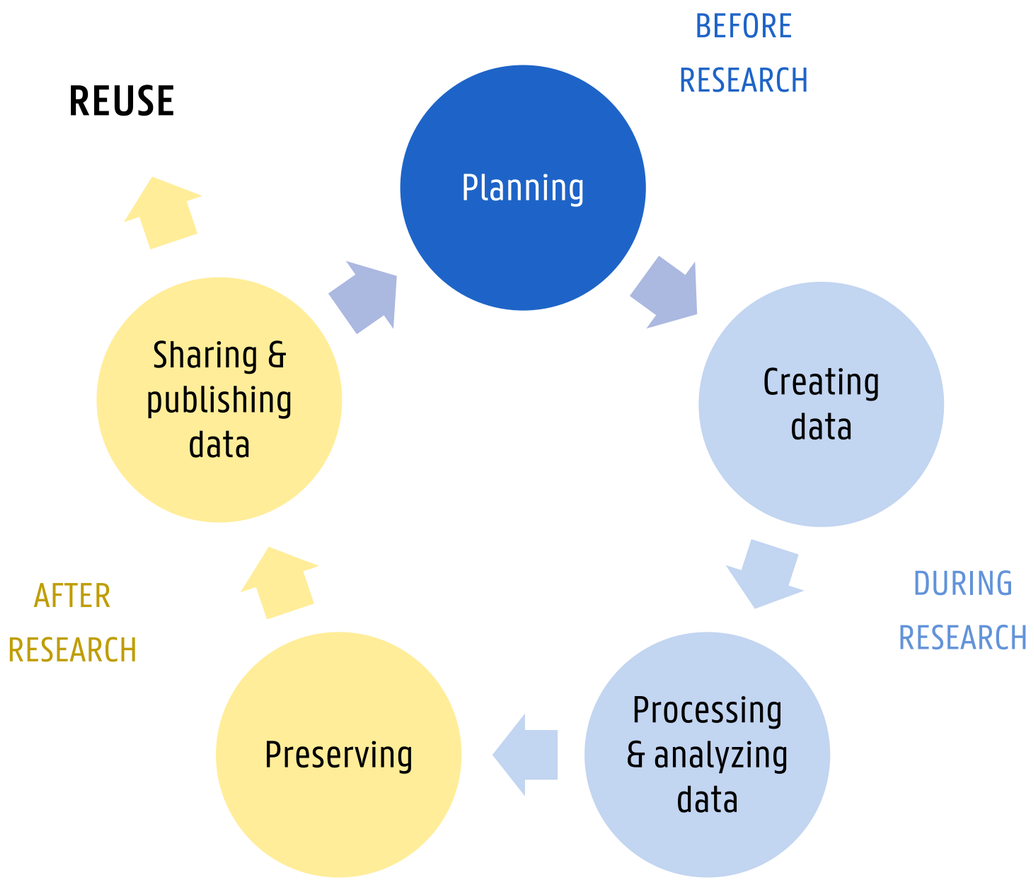


Research data is **any information** that has been collected, observed, generated or created **to validate original research findings**.

GHENT UNIVERSITY

DATA ARE FIRST-CLASS RESEARCH OUTPUT

— **Research data lifecycle:** RDM does not stop after publishing your results





PUBLICATIONS AND DATA

'Publications and Data' by Auke Herrema, licensed under CC-BY 4.0

GHENT UNIVERSITY

RDM REQUIREMENTS

GHENT
UNIVERSITY

CHANGING LEGAL & POLICY LANDSCAPE

Growing expectations of open & FAIR data

2014
- Start of H2020 Open Research Data pilot
- PLOS data availability policy

2015
- First announcement of European Open Science Cloud

2016
- UGent Policy Framework on RDM

2017
- Revised European Code of Conduct for Research Integrity (ALLEA)

2018
- FWO policy on research data management

2019
- EU Open Data Directive includes research data
- Revised BOF regulation includes data as research output

2020
- UGent Education & Examination code DMPs for starting PhDs

2021
- Horizon Europe Mandatory and recommended OS and RDM practices

2022
- Open Science progress monitoring in Flanders
- Use of Flemish Standard DMP for FWO/VLAIO-cSBO

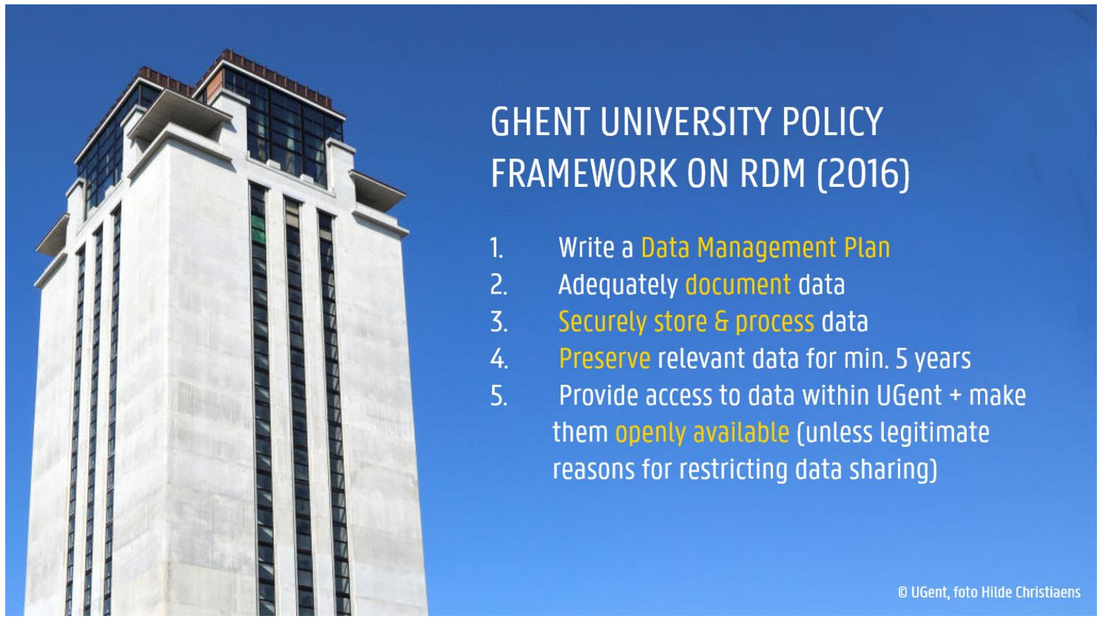
2023
- Data registration in Biblio

GHENT UNIVERSITY

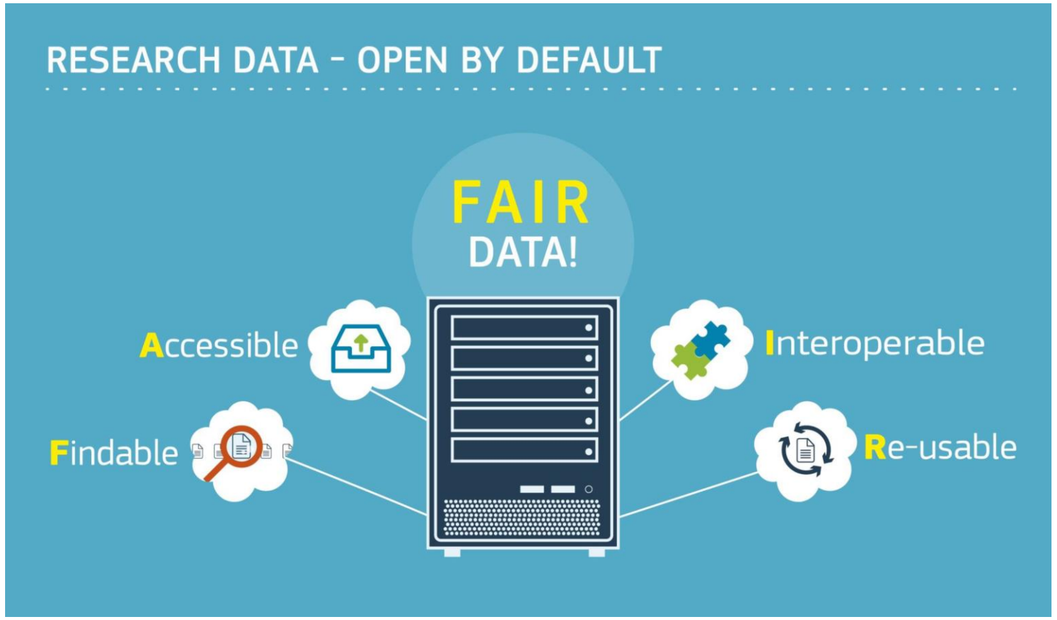
https://www.ugent.be/en/research/datamanagement/policies/overview.htm

POLICIES AND REQUIREMENTS ON DIFFERENT LEVELS

UGent RDM policy framework



HEU Open Science requirements



Journals & publishers

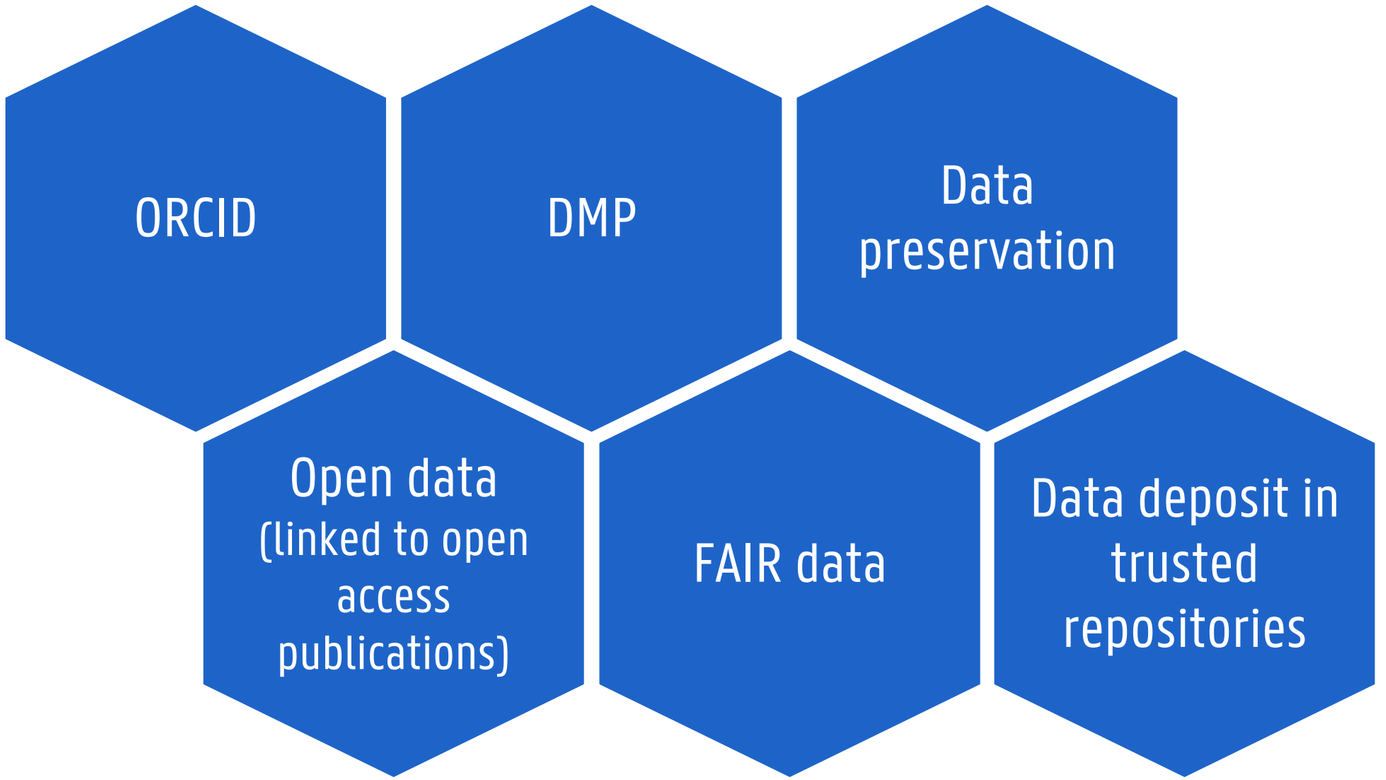
"*PLOS journals* ***require authors to make all data necessary to replicate their study's findings publicly available without restriction at the time of publication****. When specific legal or ethical restrictions prohibit public sharing of a data set, authors must indicate how others may obtain access to the data.*" PLOS ONE Data availability

ALLEA European Code of Conduct for Research Integrity



- Access as open as possible, as closed as necessary
- Data as legitimate and citable products of research

GENERAL RDM REQUIREMENTS



GHENT UNIVERSITY

EXTERNAL FUNDERS & UGENT: DMP AND RDM REQUIREMENTS

DMP	Proposal
	Initial DMP
	Final DMP
	Periodic report
	Template*
Other RDM	Open Data mandate
	Preservation period

GHENT UNIVERSITY

*All templates available via DMPonline.be
More info about external funders and UGent DMP requirements for new PhDs.

EXTERNAL FUNDERS & UGENT: DMP AND RDM REQUIREMENTS

		HEurope	ERC	BELSPO	FWO	BOF / IOF (UGent)	PhD students (UGent)
DMP	Proposal	YES	NO	YES (Provisional DMP)	NO (5 RDM questions)	NO	NO
	Initial DMP	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)
	Final DMP	YES	NO	YES	YES	YES	YES (prior to public defense)
	Periodic report	YES (Grant agreement)	NO	NO	NO	NO	NO
	Template*	YES (recommended)	YES (recommended)	YES (recommended)	YES (mandatory)	NO (use any template from DMPonline.be)	NO (use any template from DMPonline.be)
Other RDM	Open Data mandate	YES	YES	YES	NO	NO	NO
	Preservation period	NO	NO	NO	YES (≥ 5y)	YES (≥ 5y)	YES (≥ 5y)

GHENT UNIVERSITY

*All templates available via DMPonline.be

More info about external funders and UGent DMP requirements for new PhDs.

WHAT TO COVER IN YOUR DMP

GHENT
UNIVERSITY

MAIN DMP SECTIONS



Data summary / description

Purpose, origin, type, format & volume of the data



Ethical and legal issues

IRB approval, IP, GDPR



Documentation and metadata

Where and how, standards, ontologies, vocabularies



Resources

Costs



Responsibilities

Owner and roles



Data sharing and reuse

Repositories, terms, contracts, licenses



Data storage during the project

Security, back-up, transfer & access



Long-term preservation

What, where, restrictions?

GHENT UNIVERSITY

MAIN DMP SECTIONS



Data summary / description

Purpose, origin, type, format & volume of the data



Ethical and legal issues

IRB approval, IP, GDPR



Documentation and metadata

Where and how, standards, ontologies, vocabularies



Resources

Costs



Responsibilities

Owner and roles



Data sharing and reuse

Repositories, terms, contracts, licenses



Data storage during the project

Security, back-up, transfer & access



Long-term preservation

What, where, restrictions?

GHENT UNIVERSITY

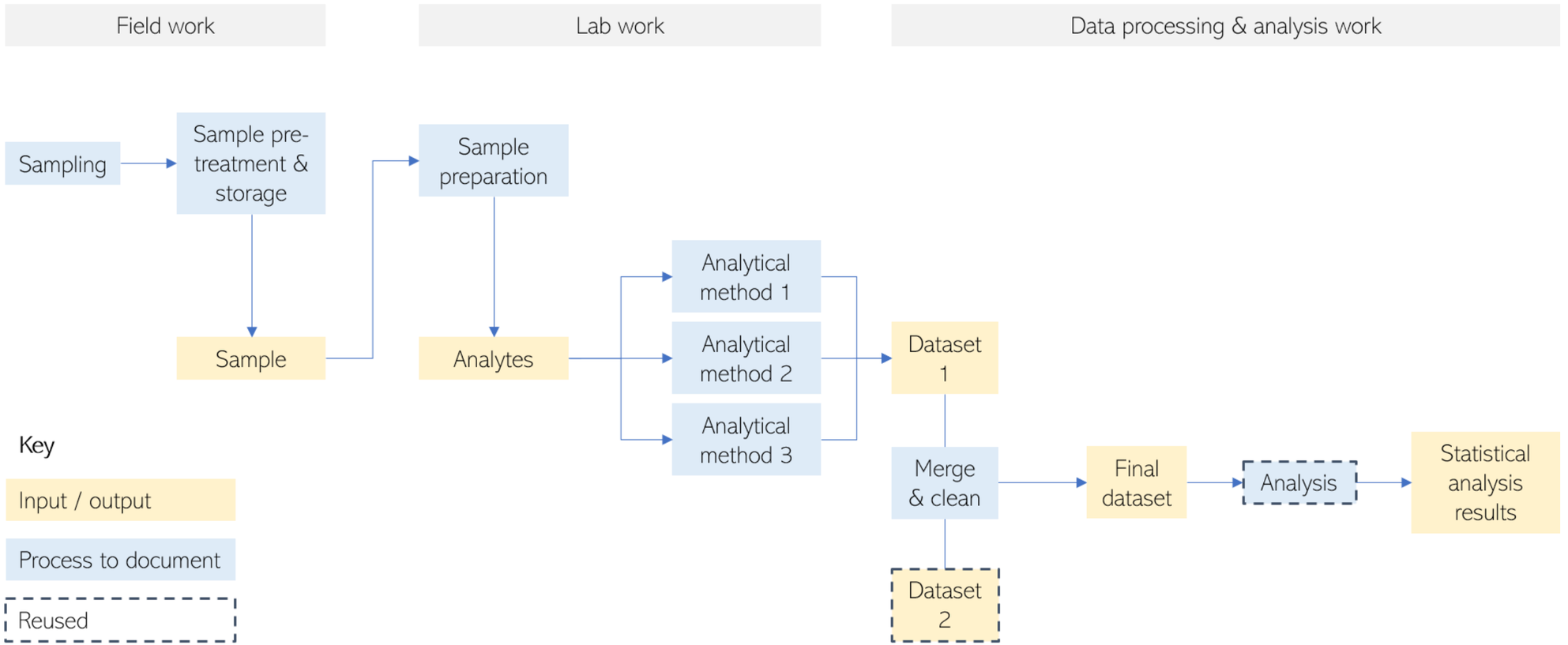
DATA DESCRIPTION – SOME TIPS

- Only provide answers relating to research data, not publications!
- Remember that 'research data' come in many different forms
- Clear distinction between re-used and new data.
- Break down and list your data types conveniently: e.g. by technique, by purpose, by research phase.
- Provide enough details for outsiders to understand the sort of data involved
  - e.g. distinguish between digital/non-digital data, quantitative/qualitative data, raw & processed data
  - specify file formats (preferably no proprietary/unusual formats)
  - specify data collection methods (e.g. experimental, observational, simulation… data)

DATA DESCRIPTION: DESIGN

Design a data flow, then describe in text or table

EXAMPLE



GHENT UNIVERSITY

DESIGN YOUR OWN DATA FLOW

- Paper and pen
- When finished, do a quick walk-through for your neighbour
  - > any missing steps? (do not review the research design ☺)

GHENT UNIVERSITY

DATA TABLE(1)

Bonus: Detect RDM needs

There is not a "one size fits all" approach

How much granularity is needed and how do I break down the data into different rows?

- It depends on your project and a good approach is to categorize the content in a way that proves useful for the management of your data
- E.g. Divide your data according to:
  - content type (quantitative vs. qualitative)
  - data collection method (observational, experimental, simulation)
  - by purpose
  - by format
  - differentiate between own data and third-party data

DATA TABLE(2)

Bonus: Detect RDM needs

There is not a "one size fits all" approach

How many columns or characteristics should I provide?

- DMP templates will explicitly require some characteristics, e.g. description, format and expected size
- Add other relevant characteristics that are specific to your data and that have implications for data management.

GHENT UNIVERSITY

CREATE YOUR DATA TABLE

- https://osf.io/2ap5t
- Download a copy of the table
- Fill out the table, ignore the ‘checklist’ column for now

GHENT
UNIVERSITY

EXAMPLE

Dataset name	Description	New or reused	Data type	Data format	Volume
Clinical data	Clinical information of cancer patients: baseline demographic information, tumor characteristics, treatment	New	Observational	eCRF REDCap database files	< 1GB
DNA- and RNA-seq data	cfDNA and extracellular RNA sequencing data obtained from plasma samples of cancer patients	New	Experimental	.bcl , .fastq , .bam , .vcf	< 5TB
ddPCR data	Digital droplet PCR data generated by Bio-Rad QX100 Droplet Reader on plasma samples of cancer patients	New	Experimental	.qlb , .qlp , .csv	< 100GB
Lab experimental data	Data generated in the lab when preparing and analysing samples; including data of quality control steps.	New	Experimental	.xlsx , .csv , .pdf , .jpeg , other	< 100GB
Documentation files	Electronic lab notes, analysis notes and quality control documents	New	Other	.xlsx , .docx , .pptx , .pdf	< 1GB
Scripts	Data analysis scripts	New	Software	.py , .R	< 1GB

RDM CHECK

GHENT
UNIVERSITY

WHAT ARE POSSIBLE RED FLAGS FOR DATA MANAGEMENT?

- What will have an impact on the way you can manage data?
- … on what you can do with data?
- What can cause obstacles during or towards the end of the project?
- …

GHENT UNIVERSITY

CHECKLIST

- Ethical approval needed
- Privacy, processing personal data
- Clinical studies: regulations
- Large data volumes
- Third party collaborators
- Reuse of data: contracts? Licenses?
- …

TASK: Add keywords in the ‘checklist column’
to your table overview for rows that will require special attention when planning data management

GHENT UNIVERSITY

STORAGE DURING PROJECTS

GHENT
UNIVERSITY

THINGS GO WRONG

Hacking/ransomware attacks

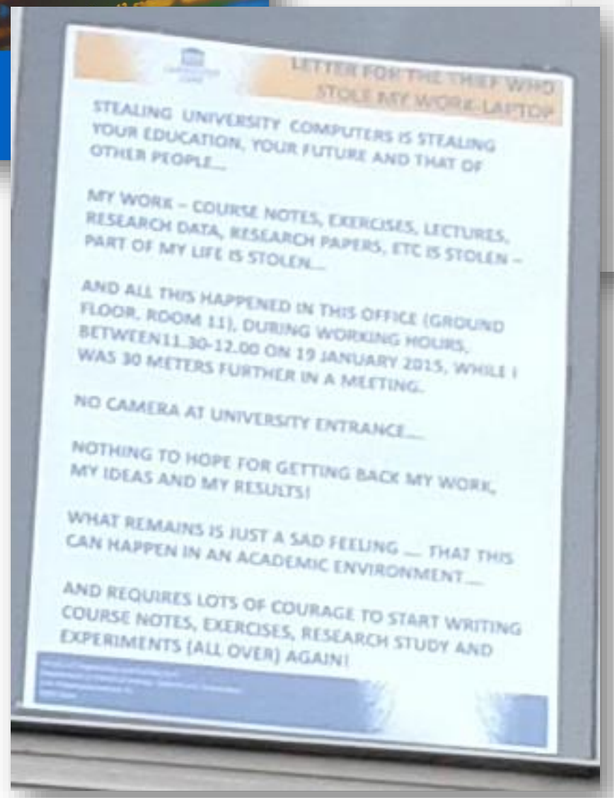
IT-veiligheids-incidenten: wat kan jij en DICT doen om deze te voorkomen

Eind juli werd er een poging tot hacking vastgesteld aan de UGent. Cybercriminelen probeerden via het Athena-platform toegang te krijgen tot de centrale infrastructuur. DICT kon dit voorkomen, maar merkt dat universiteiten steeds vaker het doelwit zijn van ransomware aanvallen en neemt daarom bijkomende maatregelen.





Inferno



Theft

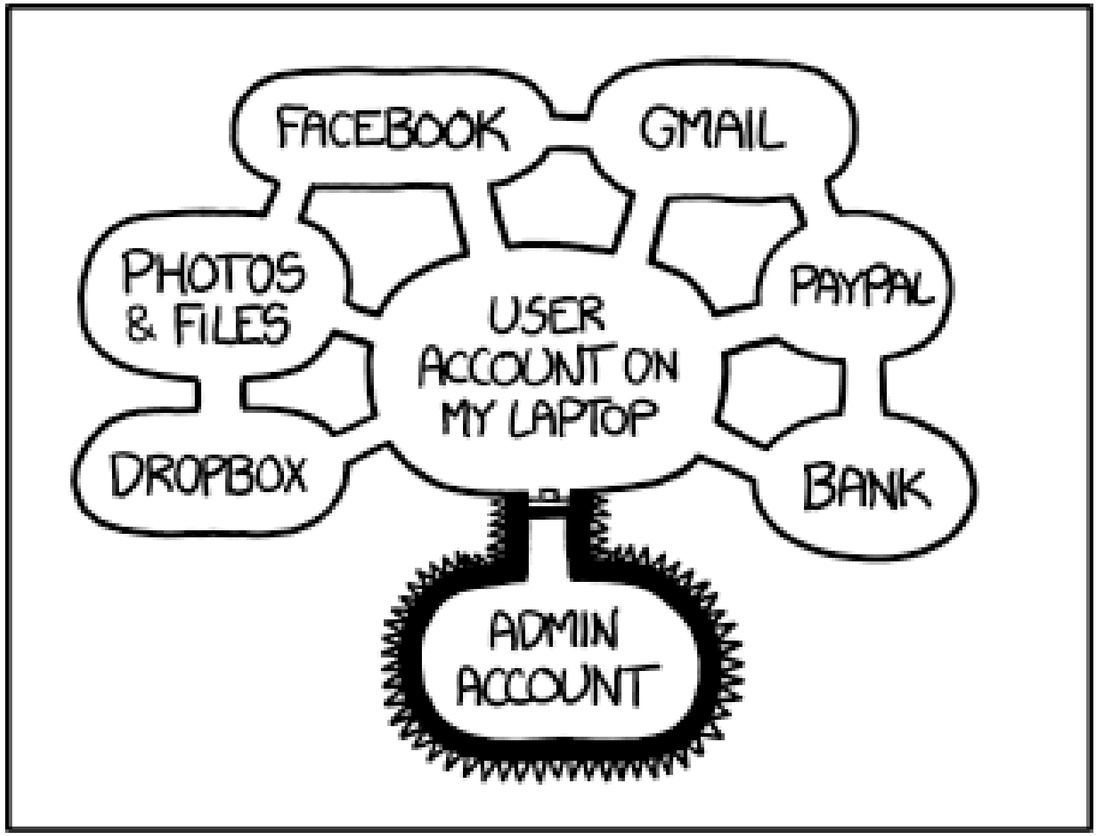


Hard disk crashes

GHENT UNIVERSITY

LOCAL STORAGE

- Any physical device you have direct access to: your PC, laptop, external drive, USB-stick, . . .
- ‘Standard’ way of working with direct access to data.
- Storing your (sensitive) data locally poses risks.
- A password does not solve the problem.
- Avoid if feasible



IF SOMEONE STEALS MY LAPTOP WHILE I'M LOGGED IN, THEY CAN READ MY EMAIL, TAKE MY MONEY, AND IMPERSONATE ME TO MY FRIENDS,

BUT AT LEAST THEY CAN'T INSTALL DRIVERS WITHOUT MY PERMISSION.

NETWORK STORAGE @ UGENT

Storage on server inside UGent

- Accessed via UGent network
- shared (aka “the share”, S:/shares/*)
- For large volumes



GHENT UNIVERSITY

MICROSOFT 365: SHAREPOINT



- **For sharing, collaborating, joint editing**
- Possible to share externally
- GB OK, TB not OK
- Access control lists
- Integration with Office
- Version control
- Access via browser: ugentbe.sharepoint.com/…
- Sync to local: accessible anywhere

GHENT UNIVERSITY

BEST PRACTICES

Where to store research data during projects?

1. Project Share
   → share with promotor
   → work directly on the shares

2. Sharepoint / Teams
   → share with promotor

! Avoid external cloud providers (Google Drive, Dropbox)

If you have to **=> Encryption!**

UZ (patient) data must remain within UZ network!

Useful guidelines:

- Requesting a project share
- Mounting network drives
- Retrieving snapshots
- ACL shares and Adjusting permissions on ACL shares

- Microsoft Teams
- Encryption guidelines

GHENT UNIVERSITY

	Microsoft 365		Shares		HPC	
Name	OneDrive for Business	Teams SharePoint Online	H:/	Project Share (acl & bulk)	HOME/DATA/ SCRATCH	VO_DATA/ VO_SCRATCH
Location	Cloud	Cloud	UGent	UGent	UGent	UGent
Size	3TB	10TB	5GB	GB-TB range	25 GB	GB-TB range
Backup mechanism	Versioning	Versioning	Snapshots	Snapshots	None	None
Sync to local	✓	✓				
Confidential	✓*	✓*	✓	✓	✓	✓
Intended use	Personal	Sharing	Personal	Sharing	Personal	Sharing
Sharing external users?	✓	✓				✓
Suitable for TB datasets				✓		✓
Data in use	✓	✓	✓	✓	✓	✓
Data at rest				✓		

* Depending on the data, encryption of data is advisable

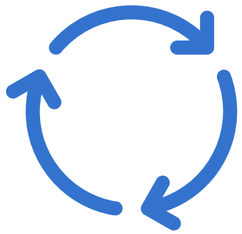
NOT ONE SIZE FITS ALL

- Multiple storage solutions might be needed.
- Other storage solutions:
  - E.g. Other (not central) institutional storage (department or group ICT)
  - E.g. multi-partner agreed storage solution
  - Get informed: what provisions are in place (e.g. back-up frequency, quota, configuration of access control…)
- Other (e.g. GitHub for code)

CONSIDERING CONFIDENTIALITY

Data for which the unwanted disclosure will have negative or harmful consequences*

(may be unlawful, put participants at risk, put business processes at risk)…

RISKS  MEASURES

GHENT UNIVERSITY

BREAK

GHENT
UNIVERSITY

DATA SHARING

GHENT
UNIVERSITY

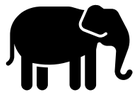
RESTRICTIONS ON DATA/SOFTWARE SHARING



Personal data



Confidential data



Sensitive data



Third-party data



Copyright



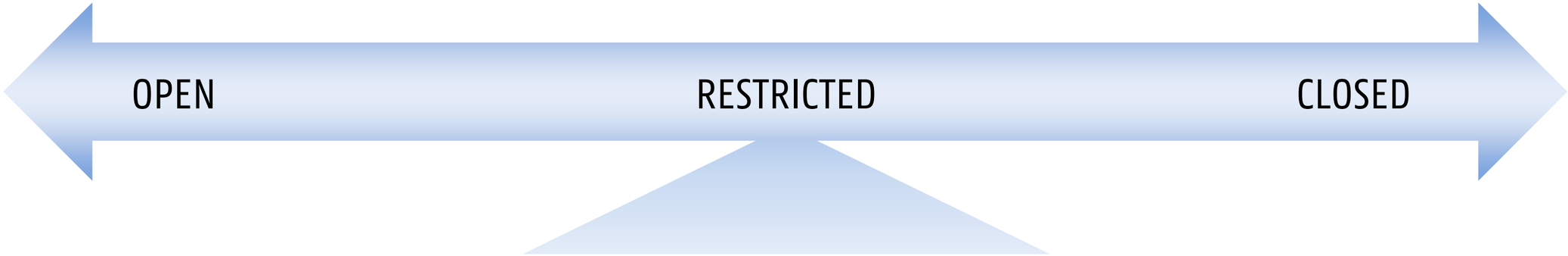
Valorization

GHENT UNIVERSITY

More info at: https://www.ugent.be/en/research/datamanagement/after-research/sharing.htm#Restrictionsondatasharing

WHAT ABOUT SHARING OF SENSITIVE OR CONFIDENTIAL DATA?

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

- Personal data
- Confidential data
- Sensitive data
- 3rd party data
- Copyright/database right
- Valorization potential

Under embargo
Unable to share

QUESTION: where does your data fit?

GHENT UNIVERSITY

PRACTICAL: WAYS OF SHARING DATA



Upon request



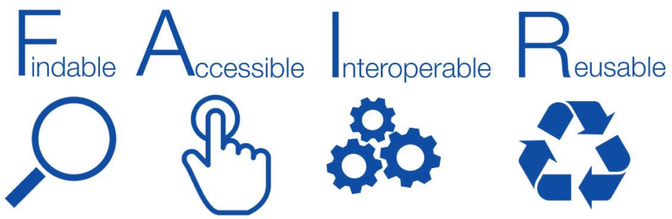
Personal or project website



Article suppl. material



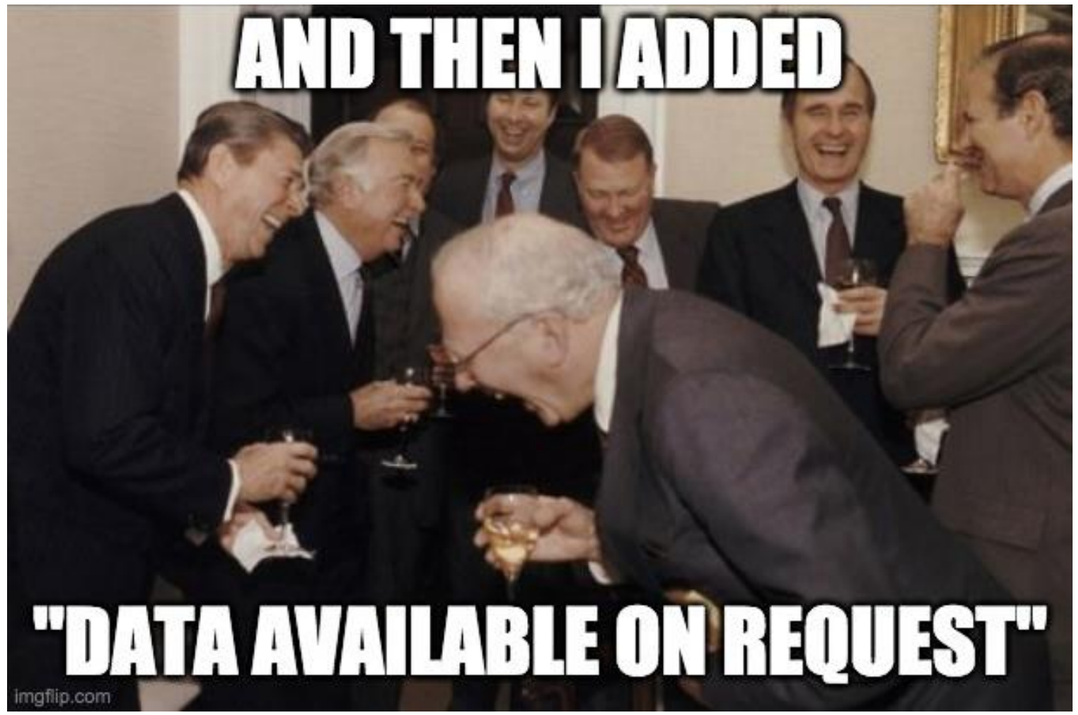
Data repository or archive



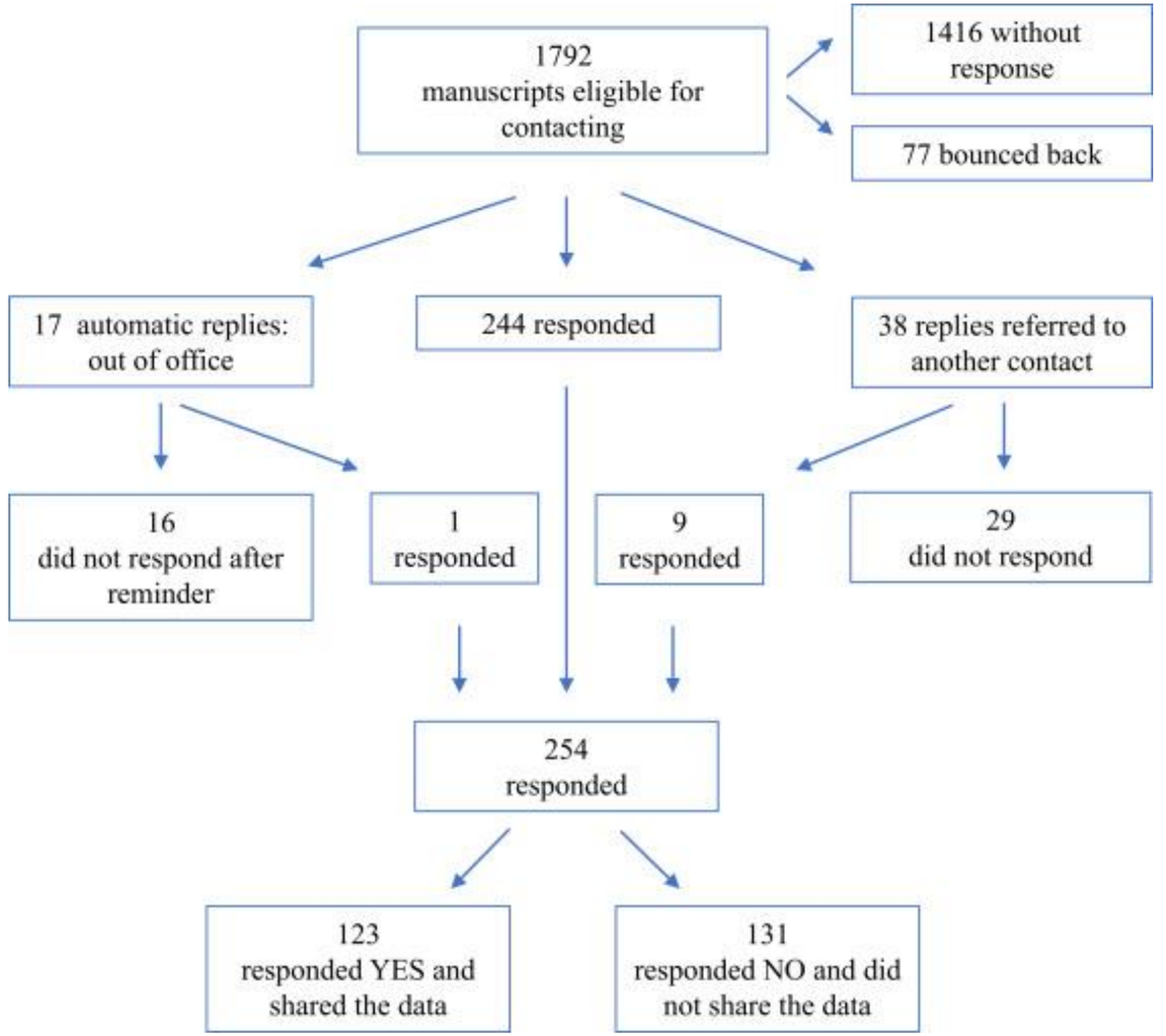
GHENT UNIVERSITY

DATA UPON REQUEST:

ONLY 14% RESPONDED, ONLY 6.8% PROVIDED THE REQUESTED DATA



https://figshare.com/articles/presentation/Handson_fMRI_code_and_data_sharing/12911165/1

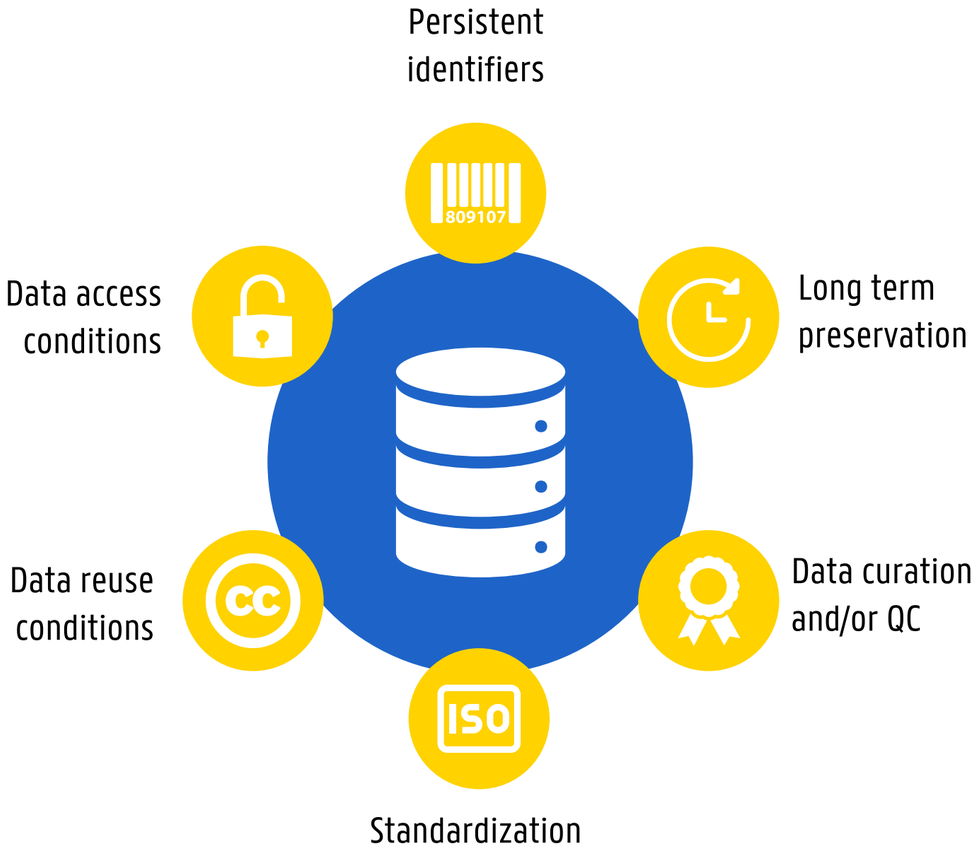


Gabelica et al. 2022, https://doi.org/10.1016/j.jclinepi.2022.05.019

GHENT UNIVERSITY

WHAT IS A DATA REPOSITORY?

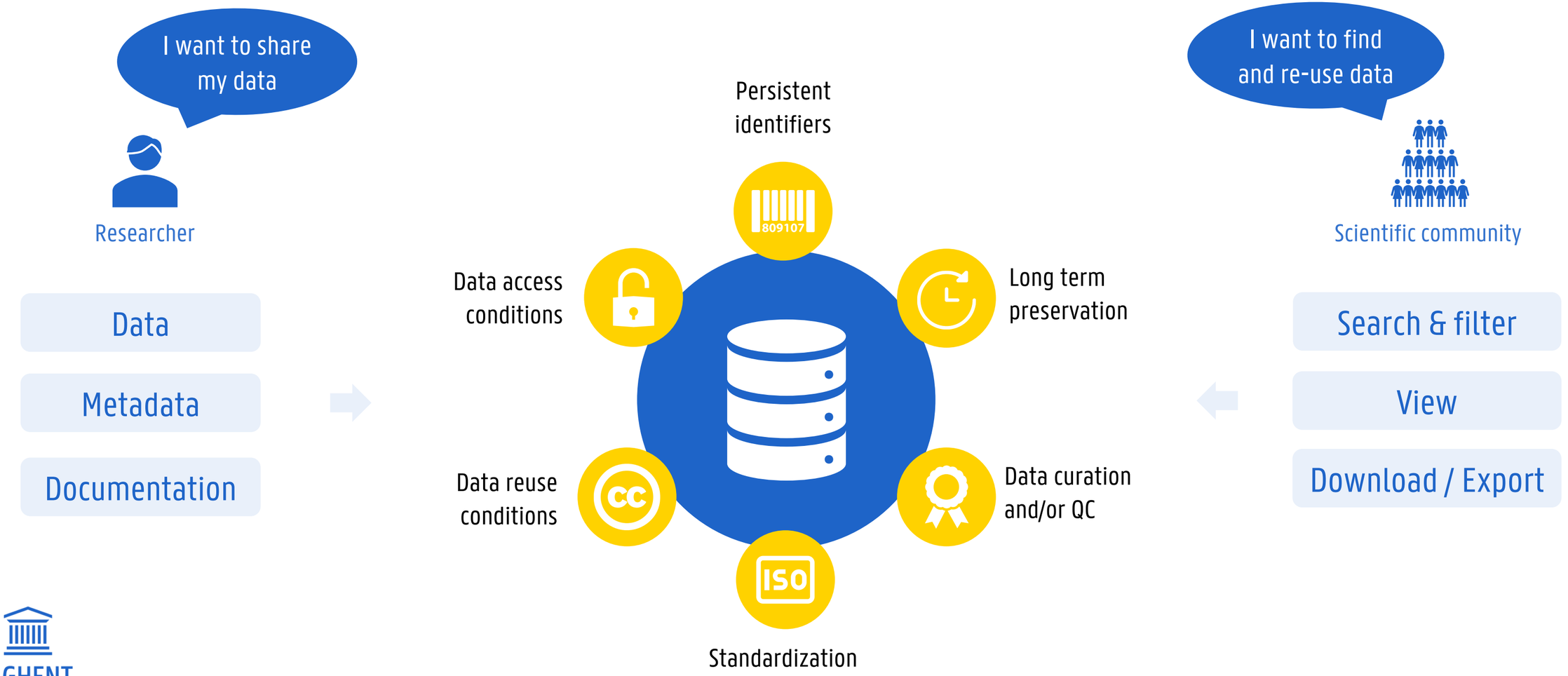
A database infrastructure that collects, manages and stores data for preservation and sharing



GHENT UNIVERSITY

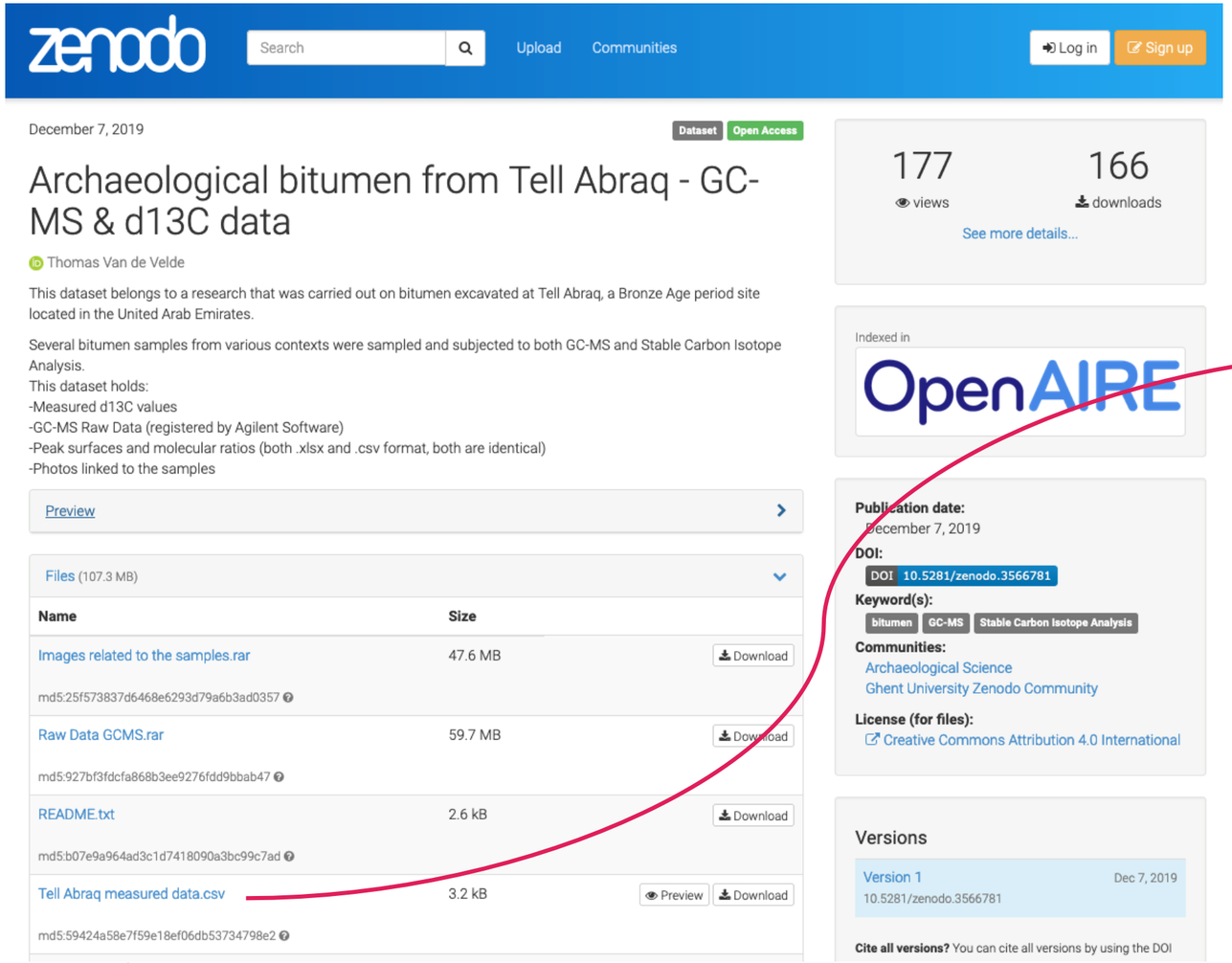
WHAT IS A DATA REPOSITORY?

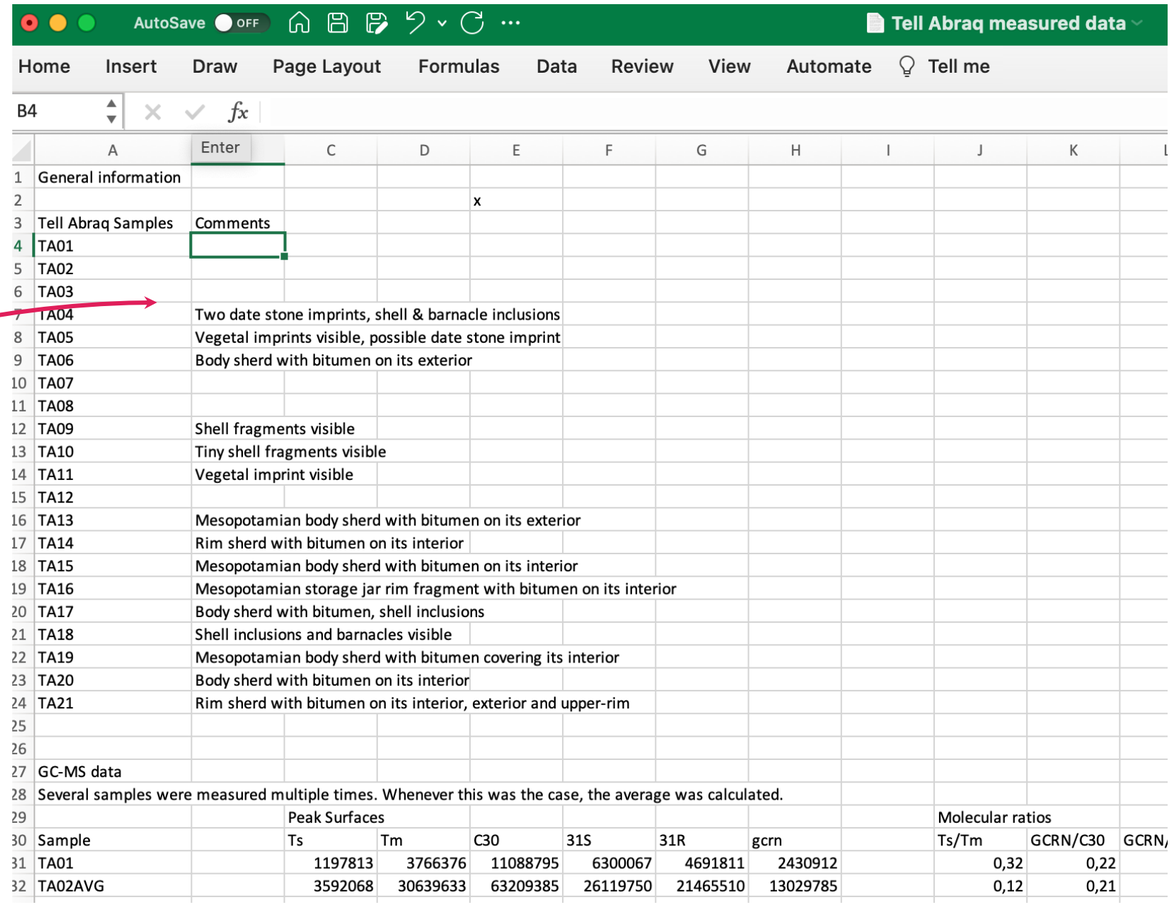
A database infrastructure that collects, manages and stores data for preservation and sharing



GHENT UNIVERSITY

DATA REPOSITORY EXAMPLE: ZENODO





	A	C	D	E	F	G	H	J	K
1	General information								
2				x					
3	Tell Abraq Samples	Comments							
4	TA01								
5	TA02								
6	TA03								
7	TA04	Two date stone imprints, shell & barnacle inclusions							
8	TA05	Vegetal imprints visible, possible date stone imprint							
9	TA06	Body sherd with bitumen on its exterior							
10	TA07								
11	TA08								
12	TA09	Shell fragments visible							
13	TA10	Tiny shell fragments visible							
14	TA11	Vegetal imprint visible							
15	TA12								
16	TA13	Mesopotamian body sherd with bitumen on its exterior							
17	TA14	Rim sherd with bitumen on its interior							
18	TA15	Mesopotamian body sherd with bitumen on its interior							
19	TA16	Mesopotamian storage jar rim fragment with bitumen on its interior							
20	TA17	Body sherd with bitumen, shell inclusions							
21	TA18	Shell inclusions and barnacles visible							
22	TA19	Mesopotamian body sherd with bitumen covering its interior							
23	TA20	Body sherd with bitumen on its interior							
24	TA21	Rim sherd with bitumen on its interior, exterior and upper-rim							
25									
26									
27	GC-MS data								
28	Several samples were measured multiple times. Whenever this was the case, the average was calculated.								
29		Peak Surfaces						Molecular ratios	
30	Sample	Ts	Tm	C30	31S	31R	gcrn	Ts/Tm	GCRN/C30
31	TA01	1197813	3766376	11088795	6300067	4691811	2430912	0,32	0,22
32	TA02AVG	3592068	30639633	63209385	26119750	21465510	13029785	0,12	0,21

GHENT UNIVERSITY

☞ More on how to use data repositories

DIFFERENT TYPES OF DATA REPOSITORIES

GENERIC REPOSITORY

zenodo

Ghent University Community

figshare

DRYAD

DOMAIN-SPECIFIC REPOSITORY

Sodha

Global Biodiversity Information Facility
Free and Open Access to Biodiversity Data

ads ARCHAEOLOGY DATA SERVICE

DataONE
Data Observation Network for Earth

INSTITUTIONAL REPOSITORY

Edinburgh DataShare

DataSTORRE
DataSTORRE: Stirling Online Repository for Research Data.

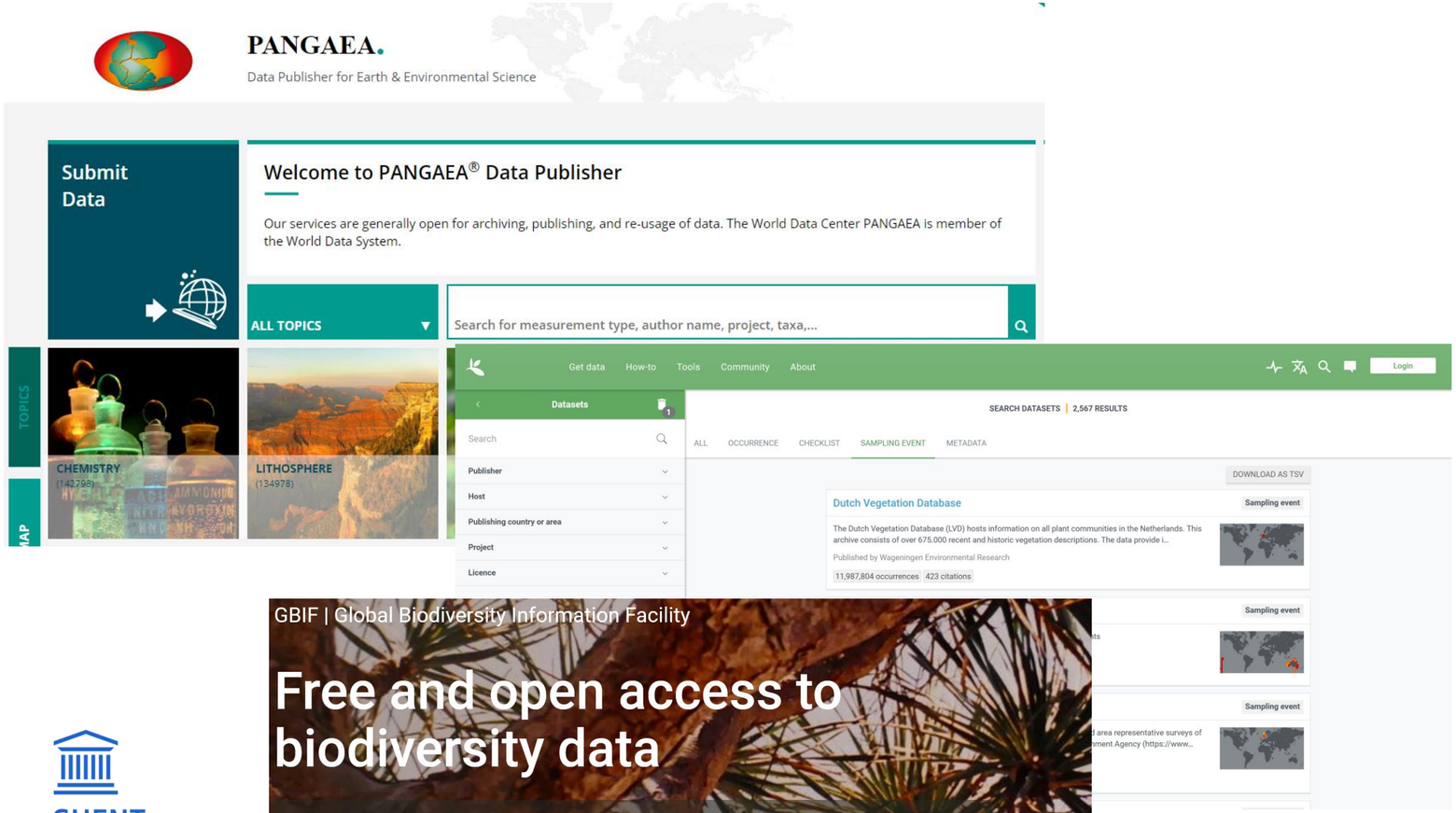
GHENT UNIVERSITY

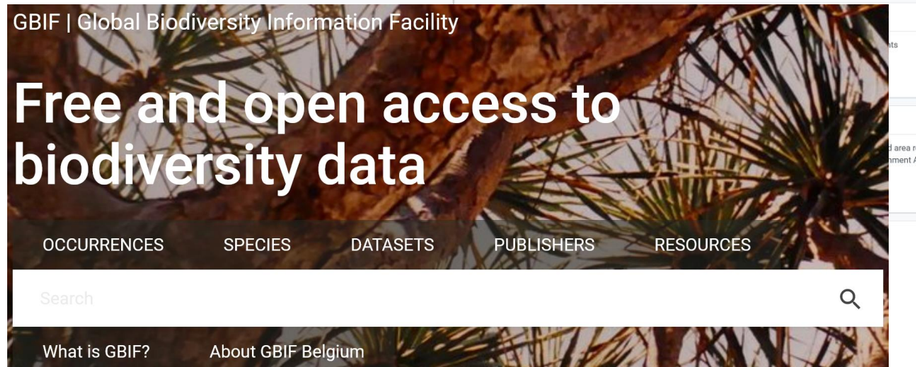
DOMAIN SPECIFIC EXAMPLES

NIH U.S. National Library of Medicine

ClinicalTrials.gov

ENA European Nucleotide Archive







GHENT UNIVERSITY

GENERIC VS. DOMAIN REPOSITORIES

- Domain-specific repositories
  - Visibility: where your peers will likely look for data
  - Use of discipline standards & vocabularies
    - Search filters, visualization tools tailored to domain
    - Interoperability: easier to integrate multiple datasets
  - Dataset curation by repository advisors often part of submission process
- Generic repositories:
  - Provide rich metadata to make it more findable & reusable
- Generic AND domain:
  - DOI, link to publication(s), dataset(s) or other outputs
  - Support minimum degree of FAIRness

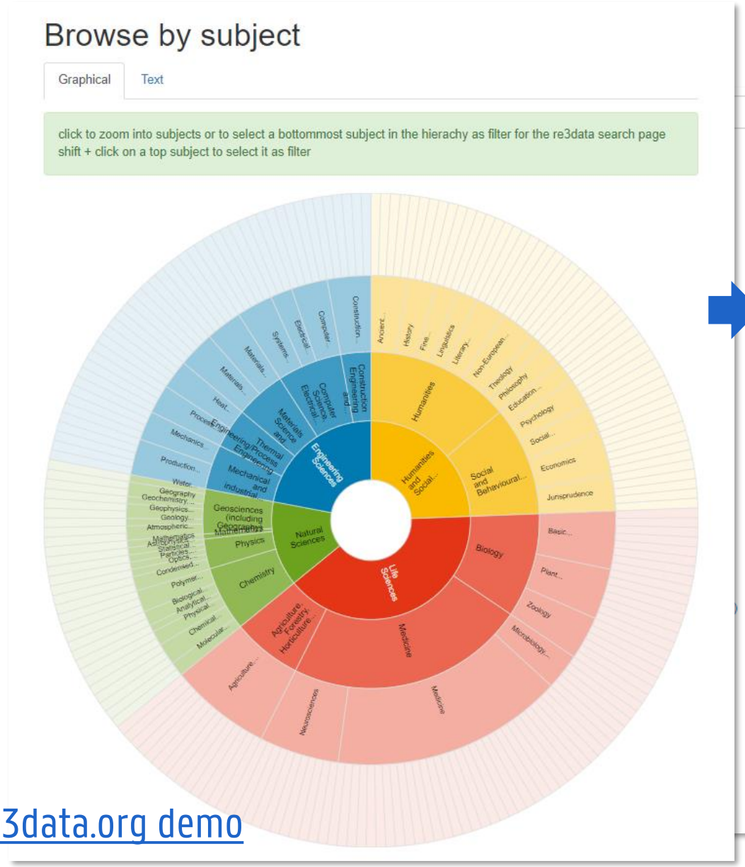
HOW WOULD YOU LOOK FOR A REPOSITORY?

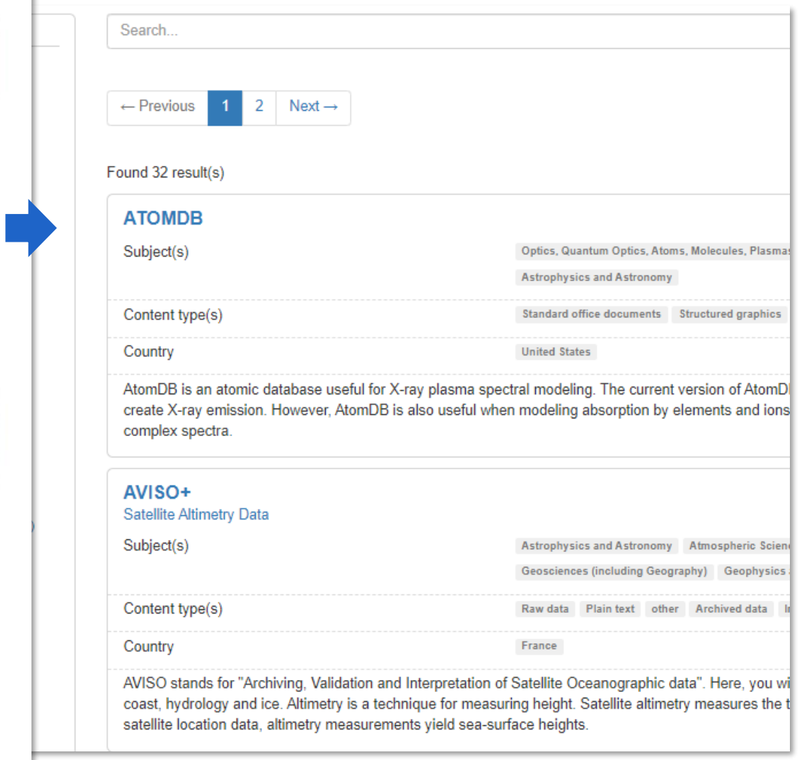
- …

GHENT UNIVERSITY

HOW TO SELECT A DATA REPOSITORY?

1. Where do your peers publish data? Domain's consensus
2. Does the journal / funder recommend or require certain repositories?
3. Registries of data repositories: re3data.org, FAIRsharing





re3data.org demo

Data Repository Guidance

Scientific Data mandates the release of datasets accompanying our Data Descriptors.

View data repositories

- **Biological sciences:** Nucleic acid sequence; Protein sequence; Molecular & supramolecular structure; Neuroscience; Omics; Taxonomy & species diversity; Mathematical & modelling resources; Cytometry and Immunology; Imaging; Organism-focused resources
- **Health sciences**
- **Chemistry and Chemical biology**
- **Earth, Environmental and Space sciences:** Broad scope Earth & environmental sciences; Astronomy & planetary sciences; Biogeochemistry and Geochemistry; Climate sciences; Ecology; Geomagnetism & Palaeomagnetism; Ocean sciences; Solid Earth sciences
- **Physics**
- **Materials science**
- **Social sciences**
- **Generalist repositories**

Data repositories recommended by Nature

FIND REPOSITORY/IES FOR YOUR DATA

- Where would your data fit?
- Find a repository
  - https://re3data.org
    - Also use this register to validate repositories you find through Google & co
  - Data repositories recommended by NaturePG
  - Biomolecular data: https://elixir-europe.org/platforms/data/elixir-deposition-databases
  - EBI Submission wizard: https://www.ebi.ac.uk/submission/
  - …
- Alternative exercise: explore Zenodo (generic repository): https://zenodo.org

DATA DOCUMENTATION & METADATA QUESTIONS



GHENT UNIVERSITY

```
project_name/
├── README.md            # overview of the project
├── data/                # data files used in the project
│   ├── README.md        # describes where data came from
│   └── sub-folder/      # may contain subdirectories
├── processed_data/      # intermediate files from the analysis
├── manuscript/          # manuscript describing the results
├── results/             # results of the analysis (data, tables, figures)
├── src/                 # contains all code in the project
│   ├── LICENSE          # license for your code
│   ├── requirements.txt # software requirements and dependencies
│   └── ...
└── doc/                 # documentation for your project
    ├── index.rst
    └── ...
```

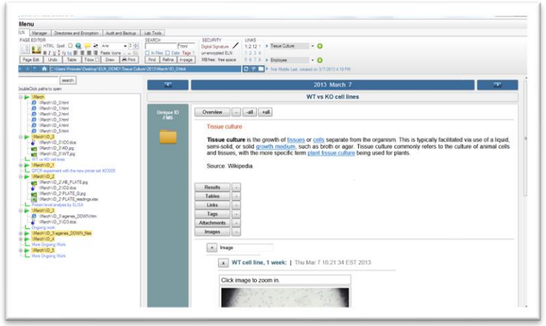
Readme files

Codebook summary table

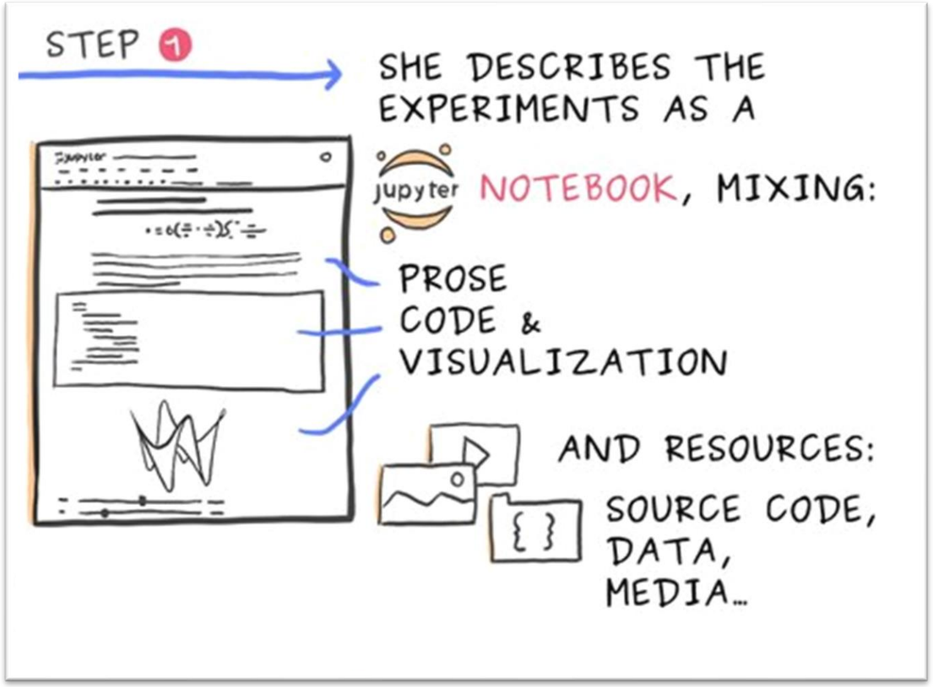
Label	Variable	Class	# unique values	Missing	Description
Sepal length in cm	**Sepal.Length**	numeric	35	0.00 %	Measured using a line gauge produced by Acme factories.
	Sepal.Width	numeric	23	0.00 %	
	Petal.Length	numeric	43	0.00 %	
	Petal.Width	numeric	22	0.00 %	
	Species	factor	3	0.00 %	Two of the three species were collected in the Gaspé Peninsula all from the same pasture, and picked on the same day and measured at the same time by the same person with the same apparatus

Codebooks





Paper and electronic lab notebooks



Jupyter notebooks

https://opendreamkit.org/2017/11/02/use-case-publishing-reproducible-notebooks/

CODEBOOK EXAMPLE

- Covid data: https://epistat.wiv-isp.be/covid/
- https://epistat.sciensano.be/COVID19BE_codebook.pdf

COVID19BE_VACC_MUNI_CUM

CUMULATIVE NUMBER OF ADMINISTERED VACCINES BY WEEK, MUNICIPALITY, AGE, AND DOSE

Variable	Definition
YEAR_WEEK	Week (Monday to Sunday) including all vaccines that were administered by the end of that week and registered in Vaccinnet+
NIS5	NIS-code (municipality) of residence of people receiving the vaccine (5 digits)
AGEGROUP	Age group of people receiving the vaccine
DOSE	Dose identifier (for vaccines requiring 2 doses: A for first dose, B for second dose; C for vaccine requiring only 1 dose; E for extra/booster dose of vaccine administered since week 36 of 2021)
CUMUL	Cumulative number of people receiving the vaccine (combinations with less than 10 vaccinations are indicated with “<10”, combinations with 0 vaccination are not shown)

EXAMPLE SEQUENCING DATA: MINSEQE

For sequencing files and derived processed files, metadata will be stored within the datafiles where possible (e.g. header of fastQ files). Additionally we will apply the MINSEQE (Minimum Information About a Next-generation Sequencing Experiment) guidelines.

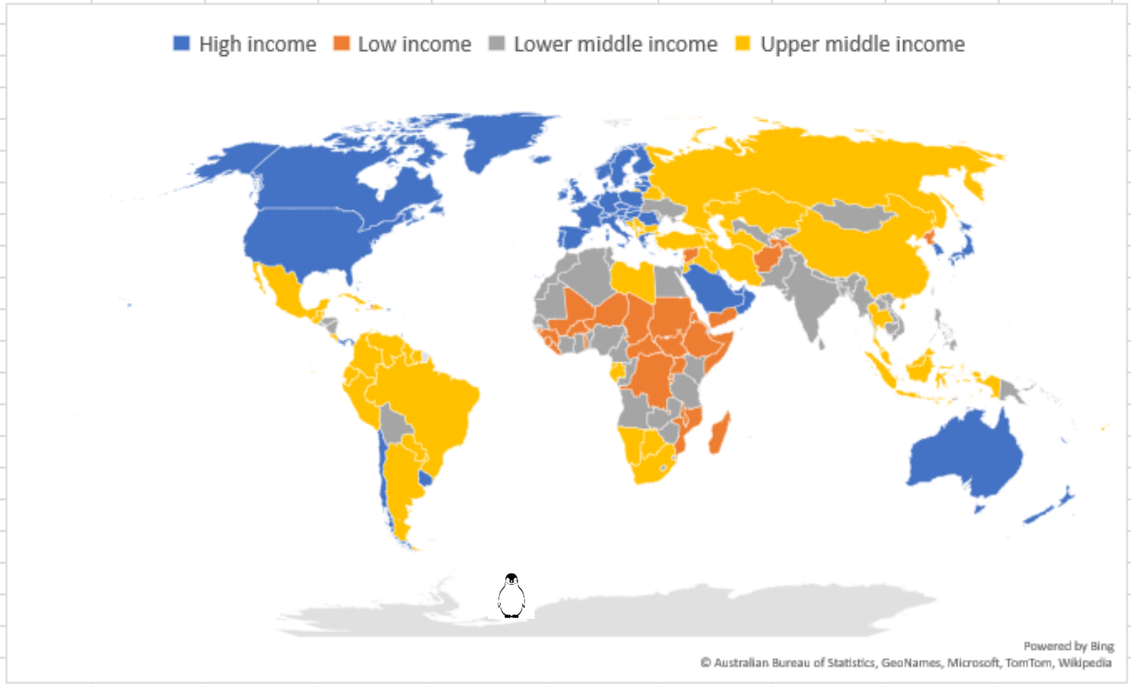
This includes:

- Raw data for each assay (e.g., FASTQ files)
- Final processed (normalized) data for the set of assays in the study (e.g., the gene expression data count matrix used to draw the conclusions in the study)
- Essential sample annotation (e.g., tissue, sex and age) and the experimental factors and their values (e.g., compound and dose in a dose response study)
- Experimental design including sample data relationships (e.g., which raw data file relates to which sample, which assays are technical, which are biological replicates)
- Sufficient annotation of the array or sequence features examined (e.g., gene identifiers, genomic coordinates)
- Essential laboratory and data processing protocols (e.g., what normalization method has been used to obtain the final processed data)

STANDARD VOCABULARIES & ONTOLOGIES

- Enable machine readability (use might be required in some domain repositories)
- Day-to-day example:

	ISOCountryCode	IncomeGroup
2	ABW	High income
3	AFG	Low income
4	AGO	Lower middle income
5	ALB	Upper middle income
6	AND	High income
7	ARE	High income
8	ARG	Upper middle income
9	ARM	Upper middle income
10	ASM	Upper middle income
11	ATG	High income
12	AUS	High income
13	AUT	High income
14	AZE	Upper middle income
15	BDI	Low income
16	BEL	High income
17	BEN	Lower middle income
18	BFA	Low income
19	BGD	Lower middle income
20	BGR	Upper middle income
21	BHR	High income
22	BHS	High income
23	BIH	Upper middle income

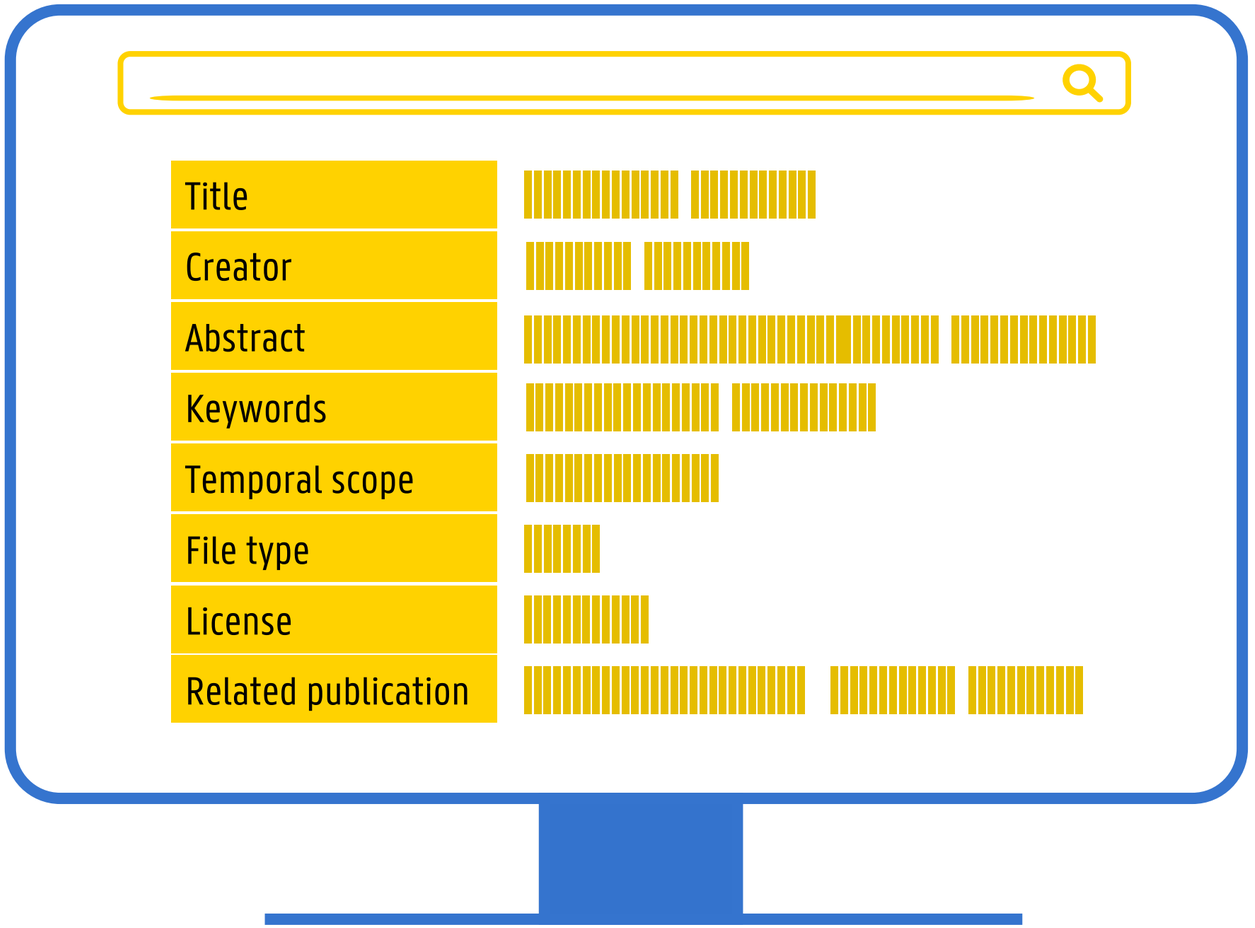


MS Excel recognizes ISO Country codes (or other place names in geonames.org) to automatically create maps

GHENT UNIVERSITY

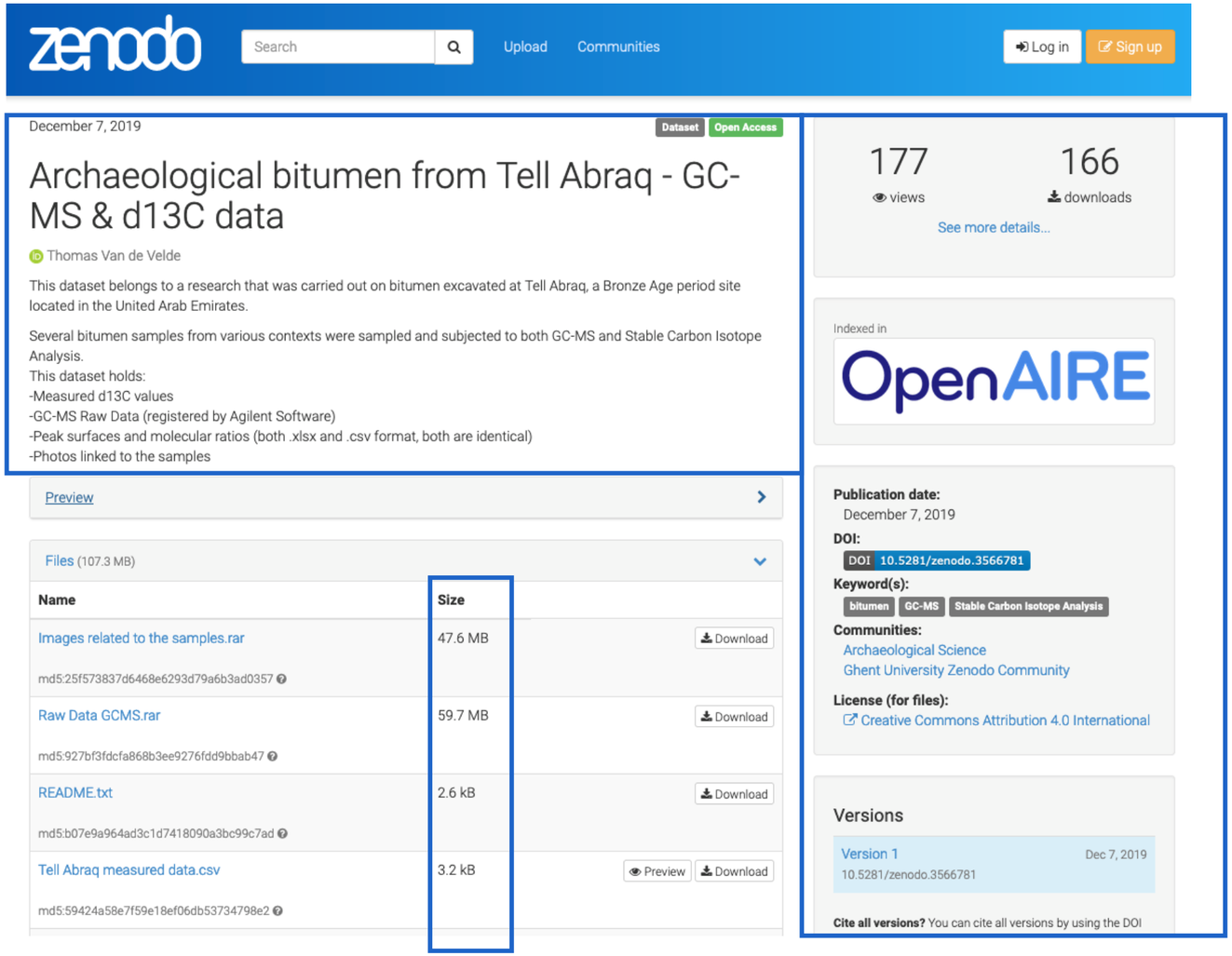
METADATA

- What metadata (= ‘data about data’) will be provided?
  - A structured, machine-readable subset of data allowing to find data
  - Will a metadata standard be used?



GHENT UNIVERSITY

DATA REPOSITORY (ZENODO): METADATA

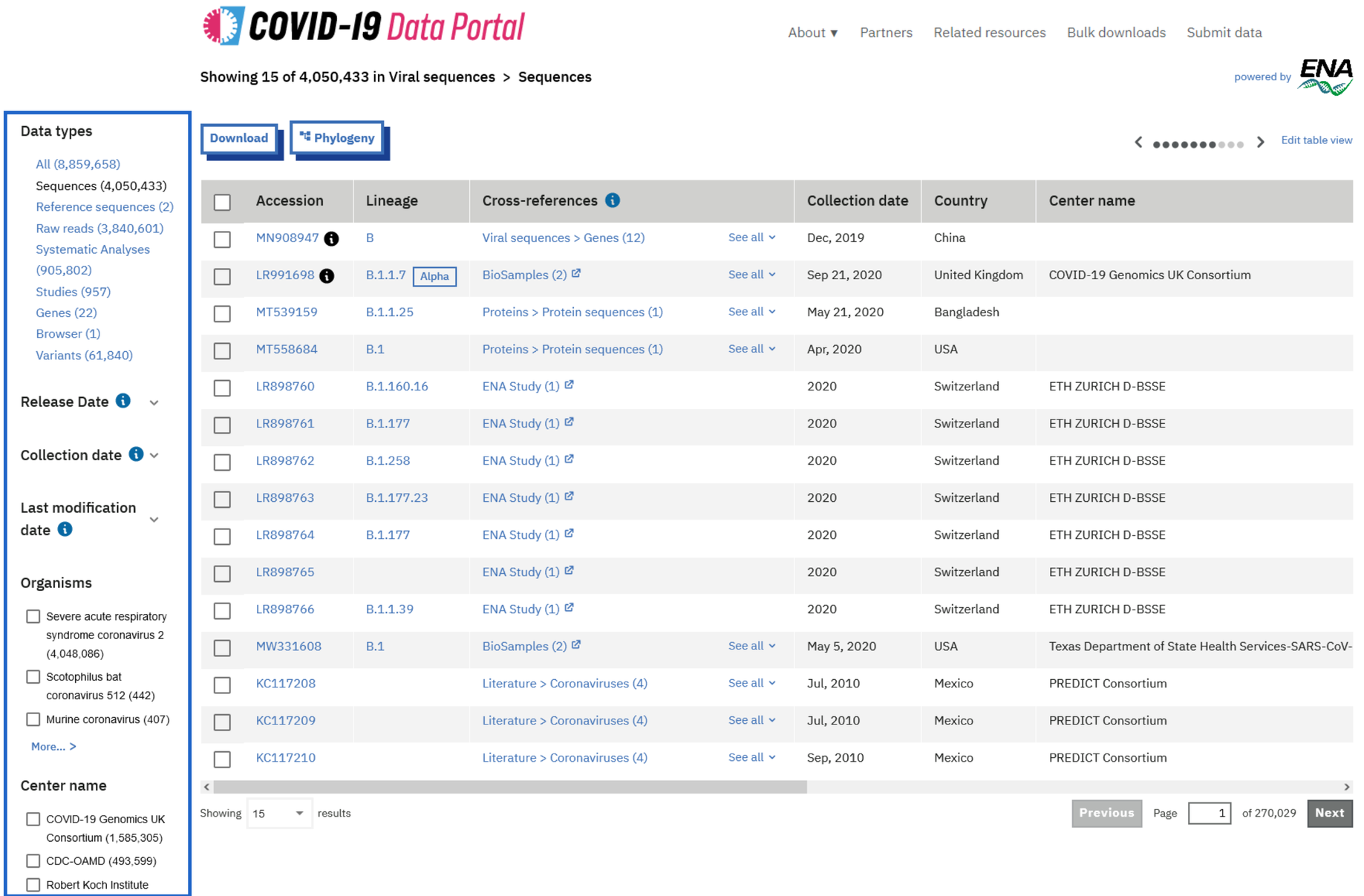


GHENT UNIVERSITY

☞ More on how to use data repositories

EXAMPLE OF A DOMAIN-SPECIFIC REPOSITORY: METADATA

COVID-19 DATA PORTAL



GHENT UNIVERSITY

METADATA: CHECK REPOSITORY SUBMISSION GUIDELINES

Example for nucleotide variation deposition: **EVA submissions**

- Prepare valid VCF file(s), which can be validated prior to submission using the EVA VCF validation suite.
- Complete a metadata template describing the samples and analyses in your study. Please provide as much metadata as possible since this information is extremely useful for downstream analysis and is directly related to the frequency at which datasets archived at EVA are reused. For reference, here is an **example** of a completed metadata template.

GHENT UNIVERSITY

DMP SOFTWARE

DMPONLINE.BE

GHENT
UNIVERSITY

DMPONLINE.BE TUTORIAL

https://onderzoektips.ugent.be/en/tips/00001281/



GHENT
UNIVERSITY

https://www.youtube.com/watch?v=5U9_DDdBzYk

SELECT YOUR TEMPLATE

- Create plan
- Ghent University as research organisation (also for UZ Ghent)
- Funder listed (i.e. funder has specific template) or not?
- Select from dropdown in case of multiple options

- Note: GDPR/AVG record is not a full DMP, but serves for GDPR registration (more info in this research tip): mandatory in case of processing personal information

GHENT UNIVERSITY

DMP LAST TIPS

- Be consistent throughout the document.
  - If a dataset cannot be shared, do not mention in a later paragraph that all data will be shared openly on Zenodo
- If certain information is not available yet, mention when and how you will address the issue
- Be complete for every question: mention all datatypes or outputs.
  - E.g.: For WP1 this question is not applicable, for WP 2-3 we will do this, and for WP 4-6 we will do that.

MORE INFORMATION

GHENT
UNIVERSITY

ADDITIONAL RESOURCES

- https://www.ugent.be/en/research/datamanagement
- https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm
- Disciplinary metadata standards: http://rd-alliance.github.io/metadata-directory/standards/
- Browse data repository by subject: https://www.re3data.org/browse/by-subject/
- DPO: dpo@uzgent.be privacy@ugent.be
- Hiruz: https://hiruz.be
- Data Science Institute (UZ only): https://intranet.uzgent.be/bad/2130/Paginas/default.aspx
- Research tips: https://researchtips.ugent.be/en/

GHENT UNIVERSITY

DOCUMENTATION RESOURCES

- READMEs
  - Guide to writing “readme” style metadata
  - README template
  - Documentation methods University of Bath
- Data Dictionaries
  - How to Make a Data Dictionary
  - Data Dictionary Template
- Codebooks
  - Codebook Cookbook
  - Sample Questionnaire with Coding
- Commented Code
  - Coding and Comment Style
- Lab/field Notebook
  - Guide for Taking Field Notes
- Metadata in practice for life sciences (by Elixir)
  - https://rdmkit.elixir-europe.org/metadata_management.html

GHENT UNIVERSITY

https://www.ugent.be/en/research/datamanagement/during-research/documentation.htm

VOCABULARIES IN LIFE SCIENCES

In clinical context often referred to as 'coding'

Clinical medicine: https://prsinfo.clinicaltrials.gov/definitions.html

Clinical ontologies in REDCap: http://bioportal.bioontology.org/

Clinical established vocabularies or ontologies

e.g. as listed in https://www.ebi.ac.uk/ols/index

or agreed terminology such as ICD-10, SNOMED, …

Bioassay ontology, eg for small molecule screening

https://bioportal.bioontology.org/ontologies/BAO

The EGA recommends using the Experimental Factor Ontology Database for describing your sample phenotypes and MONDO for disease classification

Example dataset on EGA: https://ega-archive.org/datasets/EGAD00001000002

UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

www.ugent.be

GHENT
UNIVERSITY